SCOTT, COX & ASSOCIATES, INC. consulting engineers • surveyors

May 14, 2021

Mr. Mark Rockwell
2801 N. 111th Street
Erie, CO 80516

Project: 21224

Dear Mr. Rockwell:

Enclosed with this letter are three copies of the Subsurface Investigation Report for the proposed three new residences to be constructed on a lot to be subdivided at 270 Holbrook Street in Erie, Colorado.

If there are any questions regarding our investigation or the report, please do not hesitate to contact us.

Sincerely,

SCOTT, COX & ASSOCIATES, INC.

By Kevin L. Hinds

Kevin L. Hinds, P.E.

Enclosures

1530 55th Street • Boulder, Colorado 80303 • (303) 444-3051 • Fax: (303) 444-3387

Project 21224

SUBSURFACE INVESTIGATION

Proposed New Residences for Subdivide Lot
270 Holbrook Street
Erie, Colorado

Prepared For:

Mark Rockwell
2801 N. 111th Street
Erie, CO 80516

May 2021

Prepared By:

Scott, Cox & Associates, Inc.
1530 55th Street
Boulder, Colorado 80303
(303) 444-3051

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Project 21224

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SUBSURFACE INVESTIGATION
PROPOSED NEW RESIDENCES
ON SUBDIVIDED LOT
270 HOLBROOK STREET
ERIE, COLORADO

PURPOSE

This report presents the results of a subsurface investigation performed April 30, 2021, at the site of proposed three new residences to be constructed on a lot to be subdivided at 270 Holbrook Street in Erie, Colorado. This investigation was made to provide engineering data for typical foundation systems for the proposed residences to be constructed at the site. A total of three (3) borings were completed during the course of the investigation. The locations of the borings are indicated on the Boring Location Map (Figure 1).

Factual data gathered during the field work is summarized in Figure 2 and Table 1, attached. The results of this investigation, our opinions that are based on this investigation and our experience in the general area are summarized in this report.

INVESTIGATION DETAILS

The field investigation consisted of drilling three (3) borings in or near the area of the proposed construction. The borings were completed with 4-inch diameter, continuous flight power augers using a truck-mounted drill rig.

The augers are utilized to bore and clean the hole to the desired sampling depth. The augers are then removed, and a 2-inch I.D. California spoon sampler is inserted to the desired testing depth. The sampler is then driven with blows of a standard 140-pound hammer falling a distance of 30 inches.

The sampler is driven a total of 12 inches or a maximum of 50 blows. The number of blows required to drive the sampler 12 inches, or a fraction thereof, constitutes the penetration test. The test is similar to the Standard Penetration Test described in ASTM D1586. This test, when properly evaluated, is a measure of the soil strength and density. The results of these tests are shown on the Graphic Boring Logs (Figure 2).

All soil samples recovered were inspected, a by the project engineer. The testing program consisted of performing the following tests, where appropriate.

Consolidation/Swell

- Consolidation/Swell tests were performed to determine the relative stability of the different subsurface soil types.

Natural Dry Density

- The dry density of the soils provides us with an indication of the relative compaction of the surficial soils.

Natural Moisture Content

- The moisture content test provides us with information, which may indicate the probability of instability due to consolidation or swell that may be caused by excessive wetting or drying.

Unconfined Compressive Strength

- The approximate unconfined compressive strength was determined by use of a calibrated hand penetrometer. The unconfined compressive strength can be useful in determining the bearing capacity of a soil.

PROPOSED DEVELOPMENT

As currently planned, the existing residence and outbuildings are to be razed and the lot subdivided into three residential lots. The residences will be constructed using conventional wood framing techniques and will be supported by poured-in-place reinforced concrete foundation walls over crawl spaces or basements. The loadings are anticipated to be light to moderate with no unusual loading conditions.

If actual building plans differ from the above description, we should be notified so that our recommendations can be reviewed and revised, if necessary.

SITE CONDITIONS

At the time of the investigation, the site had an existing residence and outbuildings. The ground surface was relatively flat and had a general slope down from the south to the north. The site was bordered on the east by Holbrook Street, to the north by Anderson Street and the west by an alle, and on all but the west side by existing residences. An open space was located to the west. Vegetation on the site consisted of a moderate growth of grasses and weeds along with some trees.

SUBSOILS

An approximate 1 to 1½-foot thick layer of topsoil consisting of dark brown, silty, sand and clay with organics was encountered at the surface in all of the borings.

Brown, slightly calcareous, silty, sandy clay to clayey sand was encountered beneath the topsoil in all of the borings and extended to depths of approximately 18 to 23 feet or greater than that explored. The borings encountered brown, silty, slightly clayey to clayey sand and gravel lenses below depths of approximately 14 feet. Bedrock was encountered beneath the surficial soils in two of the borings at a depth of approximately 28 feet. The bedrock consisted of gray, silty, slightly clayey sandstone. Some of the upper portions of the bedrock may be weathered.

A detailed description of the soils encountered in this investigation is presented in the Graphic Boring Logs (Figure 2).

GROUNDWATER CONDITIONS

Groundwater was encountered during our investigation in all of the borings at depths of approximately 15 to 16 feet below the existing ground surface. We are nearing the time of the seasonal high groundwater table. Therefore, some rise of the groundwater table is still anticipated. It is not possible to forecast the seasonal high groundwater table based on short duration monitoring. The only sure method of such determination is monitoring of the water table through the spring and early summer (typical seasonal high groundwater levels occur about July 1). We recommend that the bottom of any below grade spaces be maintained at least 4 feet above the seasonal high groundwater levels. Additionally, improper drainage could result in a "perched" groundwater table. This is discussed further in the "Site Drainage Considerations" section that is included later in this report. Any nearby ditches, streams or other water features will influence the groundwater elevations at the site.

FOUNDATION RECOMMENDATIONS

Any existing fill or topsoil is not considered suitable to support any foundation loadings. The silty, sandy clay to clayey sand tested is of low expansive potential. The sand and gravel and sandstone bedrock is considered to be of very low expansive potential. However, the sandstone bedrock is deep below wet soils.

Therefore, it is our opinion that the structures can be supported on footings, either continuous spread footings or isolated pad footings, founded on the low expansive, silty, sandy clay to clayey sand soils utilizing a uniform soil bearing pressure not to exceed 1,200 PSF. The loadings should be based on the dead load plus 100% of the maximum anticipated live load. A minimum dead load pressure of 300 PSF should also be utilized in the design.

The footing lines should be carefully inspected by an engineer from our office prior to placement of the footings. All footings should be placed below any fill or topsoil or potentially moderate expansive soils. The footings should also be

placed deep enough for frost protection. Any areas of soft or loose soil, which are present at the proposed footing level, should be removed down to acceptable undisturbed soils. Footings can then be placed directly upon the acceptable soils or the excavation can be backfilled to the desired footing elevation with compacted, select granular fill placed in lifts not to exceed 9 inches in thickness and compacted to a minimum of 100% of maximum density as determined by the moisture/density relationship ASTM D698.

Foundation walls supported by footings should be designed as grade beams capable of spanning a minimum distance of 12 feet. The amount of reinforcing steel used should not be less than two No. 4 bars, both top and bottom of the foundation wall. Reinforcement should be continuous around corners. Differential settlement will be minimized by proper reinforcement of foundation walls.

SLABS-ON- GRADE

The upper level soils anticipated to be beneath the slabs-on-grade are generally considered to be of low expansive potential.

The actual amount of possible slab heave is very subjective due to variability in the soils resulting in variability in expansion and also the degree and depth of wetting beneath the slabs. Outlined below is a prediction of the possible slab movements for the general soils at this site based upon a typical maximum wetting depths of five feet, which is an average worst case scenario. There was typically one soil type at the site which could influence the slabs-on-grade. That type being the low expansive silty, sandy clay to clayey sand.

Silty, Sandy Clay to Clayey Sand, (Low expansion potential)
Approximately ½ to 2 inches

It should be noted that these potential movements are only a prediction based upon the soils tested and typical slab movements seen from similar soils and wetting conditions.

There are typically four different slab-on-grade scenarios on typical residential construction. These are interior slabs in living spaces, garage slabs, patios (and stoops) and other exterior concrete like the driveway and sidewalks. We will discuss these in separate sections, below...

Interior Living Spaces

There are basically three different scenarios for floor slab construction at the site. They are as follows from least expensive with most risk of potential movement to most expensive with least risk of movement.

1. Place slabs-on-grade on existing site soils with limited subgrade preparation. The owner should be aware that some slab damage is likely to occur, similar to the existing slab-on-grade.

2. Remove and replace a portion of existing expansive soils with a non-expansive granular fill soil. This will buffer the slabs from localized heave or settlement to an extent, but some slab damage is still likely to occur. Attached to this report, Appendix A, is a discussion of the risk reduction with different depths of removal and replacement.

3. Utilize structural floor system which would isolate it from the existing site soils.

Garage Floor Slab

Standard practice in this area is to found the garage slab on native soils, with a small depth of moisture treatment (soils compacted to a relatively high moisture content). It has become common for the garage area to be dug out to the foundation level, the foundation installed and the foundation walls constructed, with the area inside the garage space filled with compacted soils. This results in a depth of fill below the garage slab of 3 feet or greater.

As such, there are three options for the unfinished area slab, as follows...

1. Fill the unfinished area with moisture stabilized native soils. Since this is a confined space and sealed by a concrete slab, we believe that this option can be considered. This will buffer the slabs from localized heave or settlement to an extent, but some slab damage is still likely to occur. Attached to this report, Appendix A, is a discussion of the risk reduction with different depths of removal and replacement, which we consider to be applicable to moisture stabilized native soils at this site.

   The fill should be compacted in maximum 9 inch lifts at 0 to +3% from optimum moisture content and compacted to a minimum of 95% of maximum density as determined by the standard moisture/density relationship test ASTM D698 (commonly called a standard proctor test).

2. Remove and replace portion of existing expansive soils with a non-expansive granular fill soil. This will buffer the slabs from localized heave or settlement to an extent, but some slab damage is still likely to occur. Attached to this report, Appendix A, is a discussion of the risk reduction with different depths of removal and replacement.

3. Utilize a structural floor system, which would isolate it from the existing site soils. While this is technically possible, it is very expensive due to the high slab loading due to the vehicular traffic and it is generally not done due to the high cost and the fact that some slab movement in the garage is generally tolerable.

We consider options 1 and 2 to be approximately equivalent in risk of floor slab movement and either of these options would be better than standard practice in the industry.

Patios and Stoops

Standard practice in this area is to found patios and stoops on native soils, with a small depth of moisture treatment (soils compacted to a relatively high moisture content) or designing them as structurally supported slabs on drilled piers or on haunches support by the adjacent foundation walls (primarily for stoops).

As such, it is our opinion that there are two options for these slabs, as follows...

1. Remove some thickness of the native soils and replace them with moisture stabilized native soils. There is a slightly greater risk of movement due to shink and swell of these soils with seasonal variations of the moisture content of the soils supporting the slab. This process will buffer the slabs from localized heave or settlement to an extent, but some slab damage is still likely to occur. Attached to this report, Appendix A, is a discussion of the risk reduction with different depths of removal and replacement, which we consider to be applicable to moisture stabilized native soils at this site.

   The fill should be compacted in maximum 9 inch lifts at 0 to +3% from optimum moisture content and compacted to a minimum of 95% of maximum density as determined by the standard moisture/density relationship test ASTM D698 (commonly called a standard proctor test).

2. Utilize structural floor system, which would isolate it from the existing site soils.

Driveway and Other Exterior Slabs

Standard practice in this area is to found driveways and other exterior slabs on native soils, with a small depth of moisture treatment (soils compacted to a relatively high moisture content). This has generally been acceptable, with, overall, a relatively small percentage of the slabs requiring replacement.

General Considerations

If slabs-on-grade supported by soil are utilized (native, moisture stabilized native or replaced with non-expansive materials), the following construction techniques will help to prevent secondary damage that could be caused by slab movement.

1. Separate slabs from the foundation elements with a slip joint. One method of doing this is to use two layers of tempered hardboard with a silicone lubricant between the boards. A slip joint should be used around the perimeter of the slab and adjacent to any other structural elements.

2. Moderately reinforce slabs with reinforcement continuous through interior slab joints. Slab joints must be provided to control the cracking. The floor joint grid should be designed to allow no more than 150 square feet of continuous slab.

3. Any load bearing partitions must be provided with their own foundation system and the slab separated as outlined above.

4. Provide a 2-inch minimum air space below any interior non-load bearing partition. It should be noted that we have seen slab movements in this area in excess of 2 inches, which have typically been caused by poor surface drainage causing seepage into the backfill and then into the soils supporting the slabs. In our opinion, 2 inches should be adequate as long as the surface drainage is properly maintained and controlled. Slab movements should be monitored so that the slab is not allowed to exert pressure on the bottom plate of non-load bearing partitions. If the slab moves within ¼ inch of the bottom plate, additional void space will have to be provided.

   If unsure of the proper construction methods to achieve the recommended air space we should be contacted for further recommendations.

5. Any pipes rising through the slab should be provided with flexible couplings or other means to allow substantial movement without damage to the piping. Any ducts connecting to equipment founded on the slab

should be equipped with flexible or crushable connections to allow for some slab movement.

6. Equipment and other building appurtenances constructed on the slab should be constructed so that slab movement will not cause damage.

Following the recommendations given above will not prevent movement of the floor slabs in the event that the moisture content of the soil beneath the slab changes. However, if movement occurs, the damage may have been reduced for a relatively small investment.

Prior to pouring a slab it is essential that all debris, topsoil and organic materials be removed. The slab subgrade soils should then be prepared and compacted utilizing the recommendations presented in the previous sections. It should be noted that failure to provide adequate fill compaction can result in settlement, which may cause slab damage such as cracking and tilting.

SITE DRAINAGE CONSIDERATIONS

It is essential that site grading be provided to prevent infiltration of surface water into the foundation system. The following methods of preventing this infiltration are recommended. These recommendations will also assist in preventing a "perched" groundwater table.

1. Mechanically compact all fill around the building, including the backfill. Compaction by ponding or saturation must not be permitted. The backfill should be compacted to not less than 90% of maximum density as determined by the standard moisture/density relationship ASTM D698. Note that some moisture may need to be added to the soils in order to obtain the proper compaction. Improper backfill compaction can cause settlement of exterior slabs such as walks, patios and driveways.

2. Provide an adequate grade for rapid runoff of surface water away from the structure (10 percent minimum for the first 10 feet away from the structure is recommended, 2 percent if paved).

3. A well constructed, leak-resistant series of gutters, or other roof drainage system, is recommended.

4. Discharge roof downspouts and all other water collection systems well beyond the limits of the backfill.

5. No irrigation within five feet of the foundation. Avoid heavy watering of any foundation plantings.

6. Observe and comply with any other precautions which may be indicated during design and construction.

It is our opinion that perimeter drainage systems should be installed at this site if the structures are to have below grade spaces (basement, garden level, or crawlspace). The perimeter drainage system should consist of 4-inch perforated pipe, surrounded by a minimum of 10 inches of 3/4 to 1-1/2 inch washed rock. The drains should be placed a minimum of 12 inches below the surface of the adjacent concrete slab or crawlspace level and should drain to a positive gravity discharge (surface discharge strongly recommended) or to a sump from which water can be pumped.

Attached to this report (Figure 3) is a drawing, which illustrates a recommended perimeter drain configuration for a footing foundation system. Additionally, more extensive drainage systems, such as underslab drainage systems, will be necessary if the below grade spaces are to extend within 4 feet of the seasonal high groundwater table. We are available to provide appropriate recommendations as necessary.

EARTH RETAINING STRUCTURES

At this site we recommend that the walls (including foundation walls and other grade beams) be designed using a lateral earth pressure equivalent to that developed by a fluid weighing 50 pcf plus any additional surcharge loads. Use of this value assumes that the wall will be backfilled with the site soils and that these soils will not be allowed to become saturated at any time during the life of the wall. Proper site grading and drainage, and installation of drainage systems at the base of any walls that are to retain soil above grade, will help to prevent saturation of the soils behind the walls. This value is valid for walls up to 10 feet in height.

LIMITATIONS

The borings in this investigation are believed to present a reasonably accurate knowledge of the existing subsoils. However, variations of subsoils not indicated by the borings are always possible. Therefore, we recommend that additional site-specific investigations be performed to confirm that the soils actually are as indicated by the investigation and to make final recommendations.

Identification of potential hazardous waste material or any coal mining activity beneath the site, if any, is beyond the scope of work for which the activities of this project were intended. We are not aware of any method to accurately determine the amount of radon at any site from a standard soils investigation. The use of

construction technology to easily retrofit a radon system, if needed, could be implemented, and is also beyond our scope of work.

It should be noted that the foundation system recommendations in this report are in accordance with the normal standard of practice assuming that the drainage recommendations provided in this report are strictly adhered to. If the soil supporting the foundation becomes wetted over a substantial period of time due to poor grading and drainage (or any other cause) , it is very possible that there could be damage to the foundation system and the slabs-on-grade. It is impractical to design a foundation system on expansive clay soils where poor site grading and drainage is allowed. In many areas along the front range expansive soil layers are relatively thick and when abnormally deep wetting occurs then typical foundation systems would not be adequate.

We would like to stress that it is not possible to fully determine the seasonal groundwater table fluctuations (and, therefore, the seasonal high groundwater table) with the short duration monitoring completed during the scope of this investigation. We have presented the method necessary to do such determination in the section titled "Groundwater Conditions". It is always possible that the groundwater table could rise to unanticipated levels, due to unknown or unrecognized groundwater sources. Unanticipated groundwater levels will also impact the recommendations, contained in this report, for the perimeter drainage system type and extent, which may be inappropriate for groundwater table levels that rise to unanticipated levels.

Due to the changing nature of geotechnical engineering practices, the information and recommendations provided in this report shall only be valid for two (2) years following the date of issue. After that time, our office should be contacted to review the information presented in this report and provide updated recommendations and design criteria appropriate for the engineering methodologies used in standard practice at that time.

INSPECTION AND QUALITY CONTROL

Placement of any significant thickness of fill, particularly fill that is to remain in place beneath loaded slabs or other structural elements, should be inspected and tested in order to verify proper compaction is being obtained. The excavations should be inspected by an engineer from our office.

Sincerely,

SCOTT, COX & ASSOCIATES, INC.

COLORADO REGISTERED
KEVIN L. HINDS
30248
5/14/21
PROFESSIONAL ENGINEER

By: _____
Kevin L. Hinds, P.E.

Reviewed

By: _____
M. Edward Glassgow, P.E.

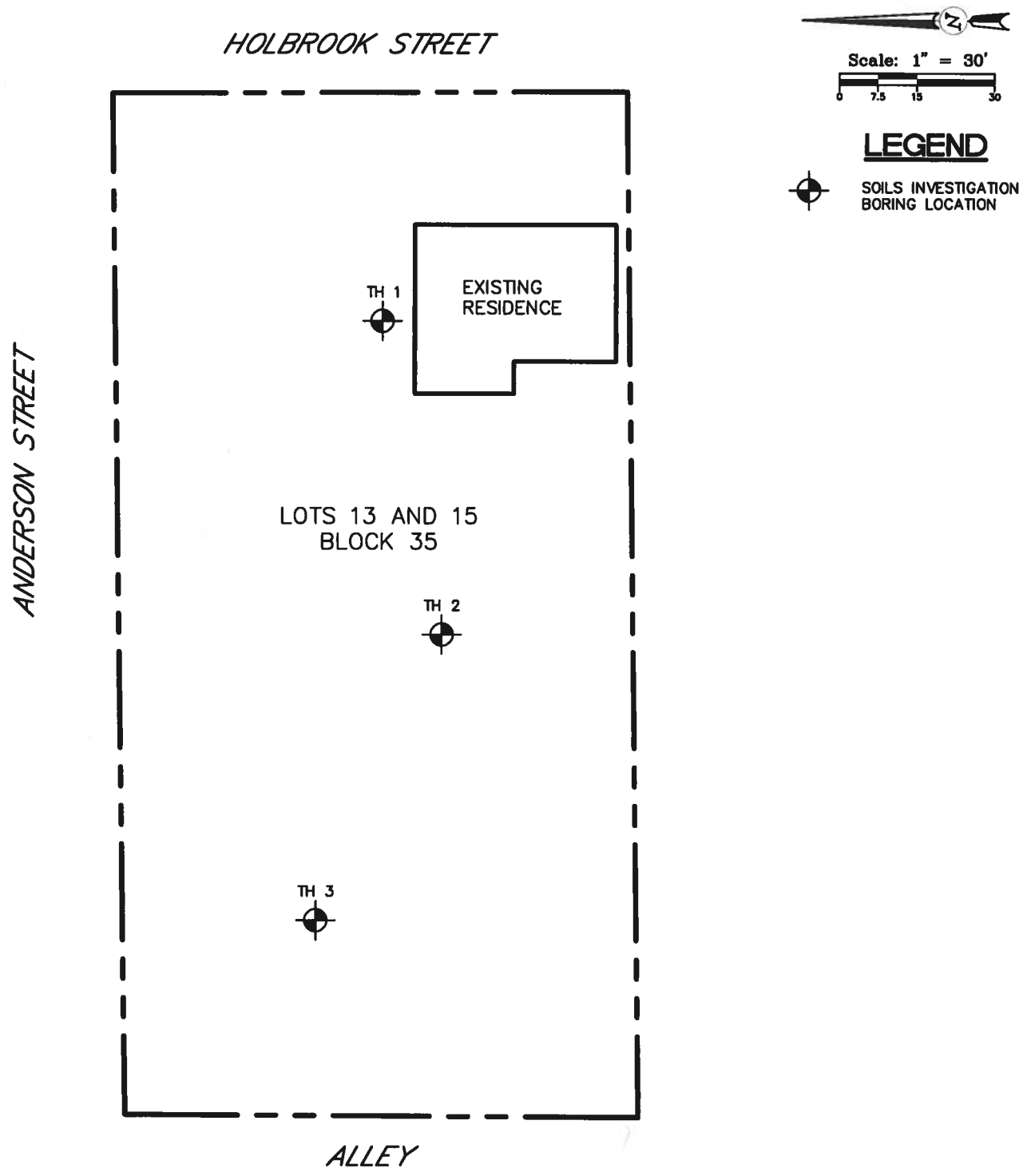

FIGURE 1
BORING LOCATION MAP

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consulting engineers • surveyors
1530 55th Street • Boulder, Colorado 80303
(303) 444 – 3051

5/5/21 JOB NO. 21224 DRAWING NO. 21224

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Graphic Boring Logs

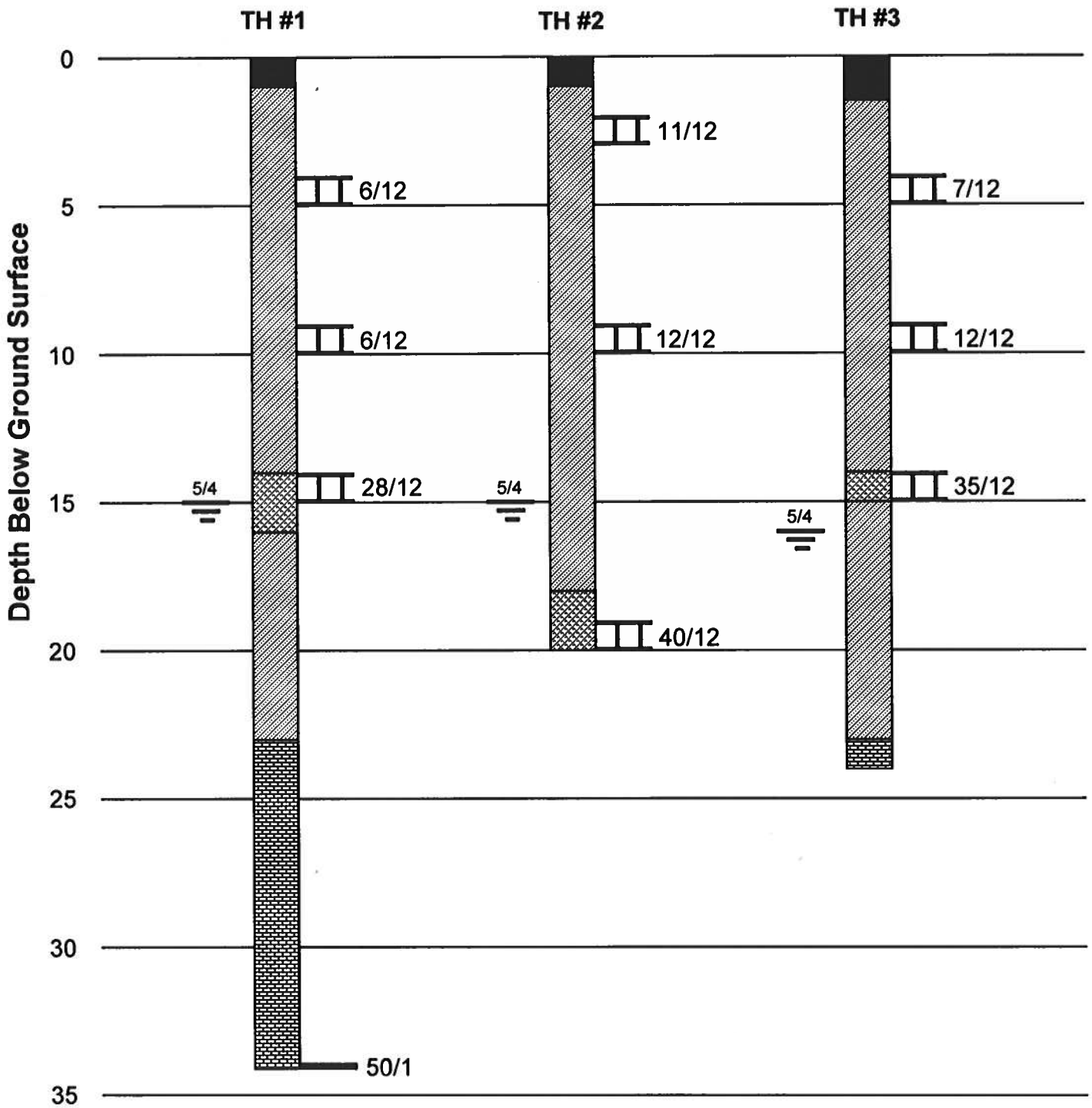

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Figure 2
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Description of Soil Types

Topsoil - Dark brown, silty, sand and clay - Contains organics

Brown, slightly calcarious, silty, sandy clay to clayey sand

Brown, silty, slightly clayey to clayey sand and gravel

Gray, silty, slightly clayey sandstone

TH #1 Soils investigation boring number

Indicates a change in soil type - May be gradual.

12/12 — 12/12 indicates that 12 blows of a 140-pound hammer falling 30 inches were required to drive a 2-inch, inside diameter sampler 12 inches.

5/4 — Indicates the groundwater table and the date that the measurement was taken

Notes

1. Borings were performed April 30, 2021 with four-inch diameter, continuous flight power augers.

2. Boring logs shown in this report are subject to the limitations, explanations and conclusions of the report.

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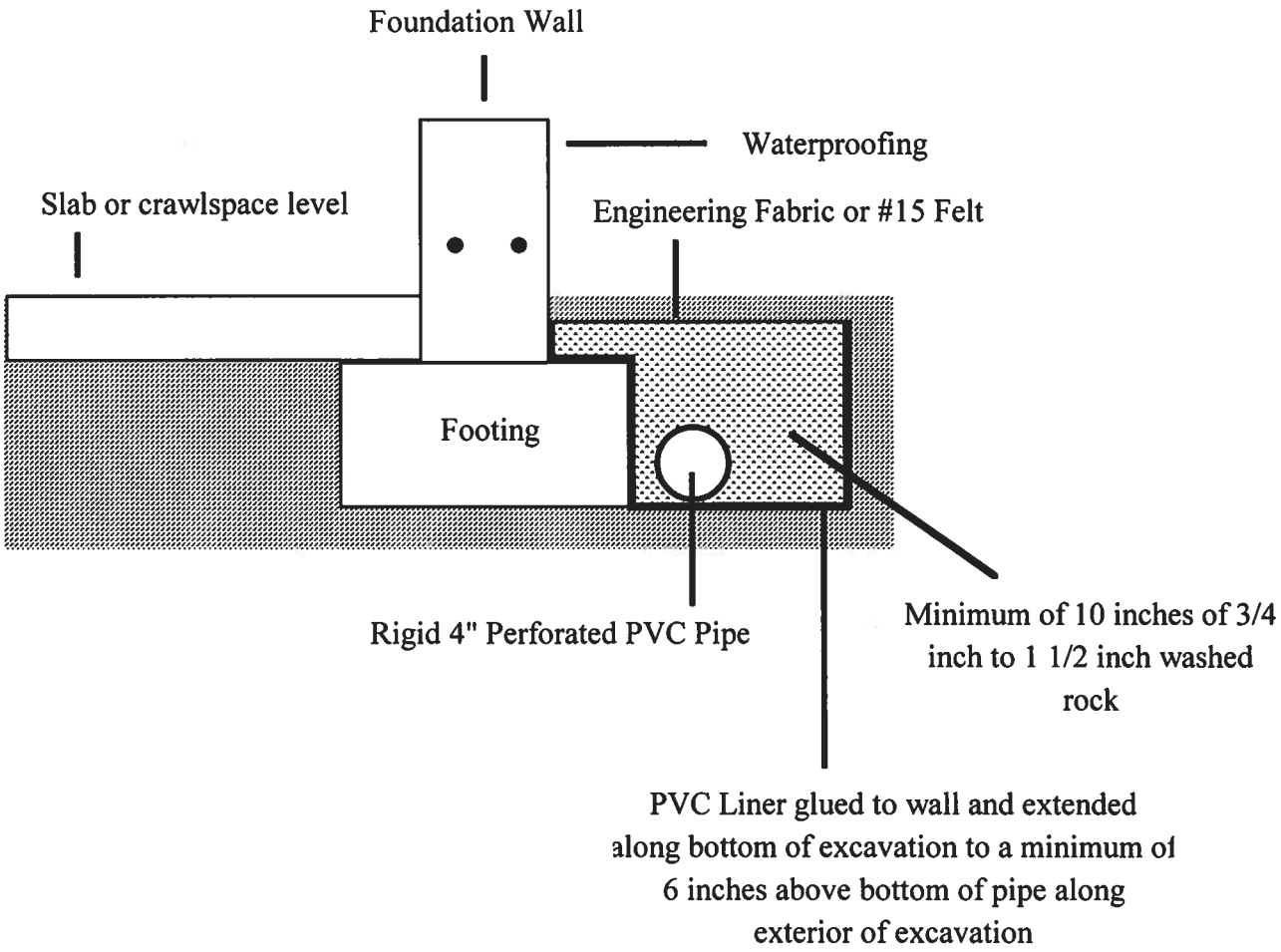
1530 55th Street • Boulder, Colorado 80303
(303) 444-3051

Typical Perimeter Drain Installation
Footing Foundation System



Notes:

1. **Slope drain and pipe at a minimum of 1/8 inch per foot to suitable outfall (sump pit or daylight outfall).**

2. **Glue all vertical T's and standpipes.**

3. **Install non-perforated pipe from perimeter pipe into sump pit.**

Figure 3

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(303) 444-3051

Table 1
Summary of Soils Properties
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PROPERTIES AT NATURAL MOISTURE CONTENT			CONSOLIDATION/SWELL				DESCRIPTION
Natural Moisture (%)	Natural Dry Density (PCF)	Unconfined Compression (PSF)	Loading (PSF)	Settlement (Dry) (%)	Settlement (Saturated) (%)	Swell (%)	
TH # 1 @ 4							
10.1	102.2	>9000	100	0.00		0.90	Brown, silty, sandy clay to clayey sand
			1000		0.30		
			2000		0.80		
0.9 % Swell upon the addition of water							
TH # 1 @ 9							
15.6	97.0	5500	100	0.60		1.10	Brown, silty, very sandy clay
			1000		0.40		
			2000		1.30		
1.7 % Swell upon the addition of water							
TH # 2 @ 2							
10.6	106.4	>9000	100	0.10		3.10	Brown, silty, sandy clay to clayey sand
			1000			0.80	
			2000		0.20		
3.2 % Swell upon the addition of water							
TH # 2 @ 9							
10.2	110.7	>9000	100	0.00		0.50	Brown, silty, sandy clay to clayey sand
			1000		0.20		
			2000		0.50		
0.5 % Swell upon the addition of water							
TH # 3 @ 4							
10.0	109.4	>9000	100	0.00		1.40	Brown, silty, sandy clay to clayey sand
			1000		0.10		
			2000		0.70		
1.4 % Swell upon the addition of water							
TH # 3 @ 9							
14.3	101.0	>9000	100	0.30		0.60	Brown, silty, sandy clay to clayey sand
			1000		0.50		
			2000		0.80		
0.9 % Swell upon the addition of water							

Appendix A
Soil Removal and Replacement Risk Analysis

An option to help mitigate or reduce slab movement is to remove and replace some of the expansive soils.

The basic concept is that sub-excavation of the expansive soils beneath the slabs or paving and replacement with suitable non-expansive materials is a method that might improve the performance of the slabs or paving. There is a relationship between the amount of material removed and the reduction of the risk. This relationship is non-linear and we consider it to be akin to a logarithmic curve with no significant risk reduction with less than 1 foot of removal and replacement and close to a 95% risk reduction with 10 feet of removal and replacement.

Therefore, we recommend that if a removal and replacement scenario is considered, that no less than 1 foot of removal and replacement be done. We anticipate that this reduces the risk by about 10% (the risk of movement being defined as the total vertical movement anticipated). The removal and replacement provides an additional benefit in that a buffer between the expansive soils and the pavement provides for moderation of the movement over a larger area. The total magnitude of the movement may be the same, but will be spread out over a larger area, which may cause less of the immediate differential cracking and heaving type of damage, which is usually associated with expansive soils.

Therefore, we summarize the risk reduction as follows:

1′ of removal and replacement	10% reduction in risk
2.5′ of removal and replacement	35% reduction in risk
5′ of removal and replacement	70% reduction in risk
7.5′ of removal and replacement	90% reduction in risk
10′ removal and replacement	95% reduction in risk

There are other factors that tend to make these values somewhat subjective. Clayey soils, such as those at this site, are very impermeable. Any non-expansive replacement soils will have permeabilities several orders of magnitude higher than the natural site soils. Therefore, water is much more easily transported through these soils. Any areas of poor grading and drainage may result in a more widespread problem than if no removal and replacement was done. Additionally, digging out an area can be analogous to digging a bathtub as there is no way for water to escape a depression that is dug and refilled with granular soils. We have seen this result in massive saturation of the soils beneath the

replacement materials, resulting in very damaging heaving, essentially defeating the entire removal and replacement scheme.

There are a number of other possible scenarios to help reduce the effects of the expansive potential of the clay soils such as chemical stabilization or moisture stabilization schemes. Chemical stabilization generally requires the mixing of either flyash, cement, lime, or other chemical into the subgrade soils for a specified depth to reduce or almost eliminate the expansion potential of the clay particles however this is only good for the depth of treatment similar to the removal and replacement as outlined above. Another method would be moisture conditioning which requires the mixture of water to the clay soils and re-compacting them generally from 1 to 3 percent above optimum moisture content. Moisture conditioning is also limited to the depth of the treatment and is also subject to problems with proper mixing of the soil and water (i.e. dry areas and wet areas) and when the clay soils are above optimum moisture content tend to pump excessively and are difficult to compact.